

A Step-Two Carnot Subcase of Ryoo's Endpoint L^1 Question

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Status

This packet records a likely valid partial solution to Question 38 / Question 42 in Ryoo's paper *Quantitative nonembeddability of groups of polynomial growth into uniformly convex spaces* [1]. The result proves that every nonabelian step-two Carnot group has a finite endpoint L^1 exponent, namely $q = 4$. It does not claim optimality in every step-two group.

Source question

Ryoo asks for which exponents $q \geq 1$ every compactly supported smooth $f : G \rightarrow \mathbb{R}$ satisfies

$$\left(\int_0^\infty \left(\int_G |f(h \exp(tv)) - f(h)| \, d\mu(h) \right)^q \frac{dt}{t^{1+q/s}} \right)^{1/q} \lesssim_{G,q} \int_G \|\nabla f(h)\|_{\ell_2^k} \, d\mu(h), \quad (1)$$

where G is a nonabelian simply connected nilpotent Lie group of step s and v is in the last nonzero lower-central term. The source notes that this is equivalent, by coarea, to a vertical perimeter versus horizontal perimeter inequality.

Question 38. Let G be a nonabelian simply connected nilpotent Lie group of nilpotency step s , and let $v \in \underbrace{[G, [G, \dots, G]]}_{s \text{ times}}$ be normalized so that $d_G(v, e_G) = 1$. For which exponents $q \geq 1$ does every smooth and compactly supported $f : G \rightarrow \mathbb{R}$ satisfy the following?

$$\left(\int_0^\infty \left(\int_G |f(h \exp(tv)) - f(h)| d\mu(h) \right)^q \frac{dt}{t^{1+q/s}} \right)^{1/q} \lesssim_{G,q} \int_G \|\nabla f(h)\|_{\ell_2^k} d\mu(h). \quad (26)$$

This tensorizes easily into an inequality for functions $f : G \rightarrow L^1$. Similarly, we may ask the following question.

Question 39. Let Γ be a not virtually abelian finitely generated group of polynomial growth. Choose v_Γ and $s > 0$ as in Theorem 7. For which exponents $q \geq 1$ do there exist $c = c(\Gamma) \in \mathbb{N}$ such that for any $n \in \mathbb{N}$ and $f : \Gamma \rightarrow L^1$ we have the following?

$$\left(\sum_{k=1}^n \frac{1}{k^{1+q/s}} \sum_{x \in B_n^\Gamma} \|f(xv_\Gamma^k) - f(x)\|_{L^1}^q \right)^{1/q} \lesssim_{\Gamma,q} \sum_{x \in B_n^\Gamma} \sum_{a \in S} \|f(xa) - f(x)\|_{L^1}.$$

By the discretization argument of Section 3, an exponent q that answers Question 38 positively also answers Question 39 positively, when G is chosen as the Malcev completion of the torsion subgroup quotient of a finite index nilpotent subgroup of Γ .

Because the case $q = \infty$ formally holds for Question 38 (see Remark 29), by Hölder's inequality the set of q that satisfies Question 38 is either of the form $(q_{G,v}, \infty)$ or $[q_{G,v}, \infty)$ for some $q_{G,v} \in [1, \infty]$ depending on G . We make the following formal definition.

Definition 40. Given a nonabelian nilpotent Lie group G , let $v \in \underbrace{[\mathfrak{g}, [\mathfrak{g}, \dots, \mathfrak{g}]]}_{s \text{ times}}$ with $d_G(\exp(v), e_G) = 1$. Let

$q_{G,v}$ denote the infimum of those $q \geq 1$ such that (26) holds for every smooth and compactly supported function $f : G \rightarrow \mathbb{R}$. Let q_G be the infimum of $q_{G,v}$ over all choices of v as above.

For $G = \mathbb{H}^3$, there is only one choice of v up to a sign, and we have $q_{\mathbb{H}^3} = 4$ with the exponent range satisfying (26) being $[4, \infty)$ [NY22]. For $G = \mathbb{H}^{2k+1}$ with $k \geq 2$, there is again only one choice of v up to a sign, and $q_{\mathbb{H}^{2k+1}} = 2$ with the exponent range satisfying (26) being $[2, \infty)$ [NY18].

For a general nilpotent Lie group G , we wish to know whether there are finite exponents q satisfying (26), since then not only would we reprove [EBGLD⁺23], but we would also have analogous quantitative nonembeddability statements: whenever q satisfies (26), then, similarly to Theorem 1, if G is Riemannian, we would have

$$c_1(B_r) \gtrsim_{G,q} (\log r)^{1/q}, \quad r \geq 2,$$

and similarly to Theorem 19, if G is Riemannian or sub-Riemannian and if N_{r_1, r_2} is an r_2 -covering of B_{r_1} , where $r_1, r_2 > 0$ with $r_1 \geq 2r_2$, and with the extra condition that $r_1, r_2 \geq 1$ if $v \in \text{span}\{X_1, \dots, X_k\}$, then

$$c_1(N_{r_1, r_2}) \gtrsim_{G,q} (\log(r_1/r_2))^{1/q}.$$

Also, if Γ is a not virtually abelian finitely generated group of polynomial growth that is quasi-isometric to G , we would have

$$c_1(B_n^\Gamma) \gtrsim_{\Gamma,q} (\log n)^{1/q}, \quad n \geq 2.$$

By the co-area formula it suffices to prove (26) when f is an indicator of a measurable set $A \subseteq G$,

Claim proved

Let G be a step-two Carnot group with Lie algebra $\mathfrak{g} = V_1 \oplus V_2$, where $V_2 = [V_1, V_1]$ is central. Equip V_1 with the fixed Carnot inner product and write $\nabla_H f$ for the horizontal gradient. For $z \in V_2$, define

$$A_z f(t) = \int_G |f(g \exp(tz)) - f(g)| dg, \quad t > 0.$$

Theorem 1. For every $v \in V_2$ there is a constant $C_{G,v} < \infty$ such that every $f \in C_c^\infty(G)$ satisfies

$$\left(\int_0^\infty \frac{A_v f(t)^4}{t^3} dt \right)^{1/4} \leq C_{G,v} \int_G |\nabla_H f(g)| dg. \quad (2)$$

Consequently, after normalizing $d_G(\exp v, e) = 1$, Question (1) has an affirmative answer with $q = 4$ for every step-two Carnot group.

Proof intuition

Every central direction in a step-two Carnot group is a finite sum of elementary commutators $[a, b]$ with $a, b \in V_1$. Each nonzero elementary commutator generates a three-dimensional Heisenberg subgroup. Naor and Young proved the exact L^4 endpoint vertical-horizontal inequality in $\mathbb{H}^3$ [2]. Applying that theorem on each left coset of the Heisenberg subgroup controls translation in the direction $[a, b]$ by the ambient horizontal gradient. Since all V_2 -directions are central in a step-two group, translating by a sum of commutators can be telescoped into translations by the individual commutators.

Supporting theorem used

Naor and Young's Theorem 1.1 says, in functional form, that in the three-dimensional Heisenberg group

$$\left(\int_0^\infty \left(\int_{\mathbb{H}^3} \frac{|F(h \exp(tZ)) - F(h)|}{\sqrt{t}} dh \right)^4 \frac{dt}{t} \right)^{1/4} \lesssim \int_{\mathbb{H}^3} (|XF(h)| + |YF(h)|) dh.$$

Equivalently,

$$\left(\int_0^\infty \frac{(\int_{\mathbb{H}^3} |F(h \exp(tZ)) - F(h)| dh)^4}{t^3} dt \right)^{1/4} \lesssim \int_{\mathbb{H}^3} (|XF(h)| + |YF(h)|) dh.$$

The relevant theorem statement is reproduced below for context.

open questions. We will then describe our main conceptual contribution, called a *foliated corona decomposition*, which is a new structural methodology that we introduce in the *proof* of this theorem; see Remark 1.2 and mainly Section 1.2 for an overview.

For a smooth function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ define $Xf, Yf: \mathbb{R}^3 \rightarrow \mathbb{R}$ by setting for $h = (x, y, z) \in \mathbb{R}^3$,

$$Xf(h) \stackrel{\text{def}}{=} \frac{\partial f}{\partial x}(h) + \frac{1}{2}y \frac{\partial f}{\partial z}(h) \quad \text{and} \quad Yf(h) \stackrel{\text{def}}{=} \frac{\partial f}{\partial y}(h) - \frac{1}{2}x \frac{\partial f}{\partial z}(h). \quad (1)$$

Also, for $t \in (0, \infty)$ define $D_v^t f: \mathbb{R}^3 \rightarrow \mathbb{R}$ by setting for $h = (x, y, z) \in \mathbb{R}^3$,

$$D_v^t(h) \stackrel{\text{def}}{=} \frac{f(x, y, z+t) - f(h)}{\sqrt{t}}. \quad (2)$$

Theorem 1.1. *Every compactly supported smooth function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ satisfies¹*

$$\left(\int_0^\infty \left(\int_{\mathbb{R}^3} |D_v^t f(h)|^4 dh \right) \frac{dt}{t} \right)^{\frac{1}{4}} \lesssim \int_{\mathbb{R}^3} (|Xf(h)| + |Yf(h)|) dh. \quad (3)$$

Moreover, one cannot replace the $L_4(\frac{dt}{t})$ norm above by an $L_q(\frac{dt}{t})$ norm for any $0 < q < 4$.

The second assertion (sharpness) of Theorem 1.1 resolves negatively the conjecture of [LN14b] that (3) holds with the $L_4(\frac{dt}{t})$ norm in the left hand side replaced by the $L_2(\frac{dt}{t})$ norm. Notwithstanding the optimality of (3), it should be noted that it was previously unknown whether such a bound holds true merely for *some* finite exponent, namely that there exists $0 < p < \infty$ such that in the setting of Theorem 1.1 we have

$$\left(\int_0^\infty \left(\int_{\mathbb{R}^3} |D_v^t f(h)|^p dh \right) \frac{dt}{t} \right)^{\frac{1}{p}} \lesssim \int_{\mathbb{R}^3} (|Xf(h)| + |Yf(h)|) dh. \quad (4)$$

It is simple to justify (see [NY18, Remark 4]) that if (4) holds, then the analogous bound holds for any larger exponent $P > p$.

Proof

We first transfer the $\mathbb{H}^3$ theorem to elementary bracket directions in G .

Lemma 1. *Let $a, b \in V_1$ and $z = [a, b] \neq 0$. Then there is a constant $C_{a,b,G}$ such that*

$$\left(\int_0^\infty \frac{A_z f(t)^4}{t^3} dt \right)^{1/4} \leq C_{a,b,G} \int_G (|af(g)| + |bf(g)|) dg \leq C'_{a,b,G} \int_G |\nabla_H f(g)| dg$$

for every $f \in C_c^\infty(G)$. Here af and bf denote the left-invariant derivatives in the horizontal directions a and b .

Proof. Let $H \leq G$ be the connected subgroup with Lie algebra $\mathfrak{h} = \text{span}\{a, b, z\}$. Since $z = [a, b] \neq 0$ and G has step two, $\mathfrak{h}$ is a three-dimensional Heisenberg Lie algebra, up to a linear rescaling of the horizontal basis and central coordinate.

Because G and H are unimodular nilpotent Lie groups, Weil integration gives a quotient measure on G/H such that

$$\int_G \Phi(g) dg = \int_{G/H} \int_H \Phi(xu) du d\bar{x}$$

for compactly supported measurable Φ . For each coset representative x , apply the Naor–Young $\mathbb{H}^3$ inequality to $F_x(u) = f(xu)$ on H . The constants change only by the fixed linear equivalence between (a, b, z) and the standard Heisenberg basis. Thus

$$\left(\int_0^\infty \frac{(\int_H |f(xu \exp(tz)) - f(xu)| du)^4}{t^3} dt \right)^{1/4} \leq C_{a,b} \int_H (|af(xu)| + |bf(xu)|) du.$$

Integrating over G/H and using Minkowski's integral inequality yields

$$\left(\int_0^\infty \frac{A_z f(t)^4}{t^3} dt \right)^{1/4} \leq C_{a,b} \int_G (|af(g)| + |bf(g)|) dg.$$

The final inequality follows because $a, b \in V_1$ are fixed horizontal vectors, so their directional derivatives are bounded by constants times $|\nabla_H f|$. $\square$

We now prove Theorem 1. Since the bracket map $\Lambda^2 V_1 \rightarrow V_2$ spans V_2 , choose $a_j, b_j \in V_1$, $1 \leq j \leq m$, such that

$$v = \sum_{j=1}^m z_j, \quad z_j = [a_j, b_j].$$

Terms with $z_j = 0$ may be omitted. Since V_2 is central,

$$\exp(tv) = \prod_{j=1}^m \exp(tz_j).$$

Let $w_0 = 0$ and $w_j = z_1 + \cdots + z_j$. For each $t > 0$,

$$\begin{aligned} A_v f(t) &= \int_G |f(g \exp(tw_m)) - f(g)| dg \\ &\leq \sum_{j=1}^m \int_G |f(g \exp(tw_j)) - f(g \exp(tw_{j-1}))| dg \\ &= \sum_{j=1}^m A_{z_j} f(t), \end{aligned}$$

where the last equality uses right-translation invariance of Haar measure by the central element $\exp(tw_{j-1})$. Applying Minkowski in $L^4((0, \infty), t^{-3} dt)$ and then Lemma 1,

$$\left(\int_0^\infty \frac{A_v f(t)^4}{t^3} dt \right)^{1/4} \leq \sum_{j=1}^m \left(\int_0^\infty \frac{A_{z_j} f(t)^4}{t^3} dt \right)^{1/4} \leq C_{G,v} \int_G |\nabla_H f| dg.$$

This proves (2).

For every $q \geq 4$, the $q = \infty$ deterministic path bound gives $\sup_{t>0} A_v f(t)/\sqrt{t} \lesssim_{G,v} \int_G |\nabla_H f|$. Combining this with (2),

$$\int_0^\infty \left(\frac{A_v f(t)}{\sqrt{t}} \right)^q \frac{dt}{t} \leq \left(\sup_{t>0} \frac{A_v f(t)}{\sqrt{t}} \right)^{q-4} \int_0^\infty \left(\frac{A_v f(t)}{\sqrt{t}} \right)^4 \frac{dt}{t},$$

so the endpoint inequality also holds for all $q \geq 4$.

Verification notes and limitations

The proof is exact for the continuous step-two Carnot subcase of Question 38 / Question 42. It is not an optimal-exponent theorem: for $\mathbb{H}^{2k+1}$, $k \geq 2$, Ryoo records the sharper known value $q = 2$, while the argument here gives only the weaker admissible exponent $q = 4$. No claim is made for higher-step Carnot groups or for arbitrary non-Carnot nilpotent Lie groups.

The main transfer point for review is Lemma 1: one should check the use of Weil disintegration over left cosets of the elementary Heisenberg subgroup and the harmless linear rescaling of Naor–Young’s standard $\mathbb{H}^3$ theorem.

The novelty check was bounded. I searched the run indexes for arXiv:2207.11305 and for “step-two Carnot”, “vertical perimeter”, “L1”, “Naor”, “Young”, and Heisenberg variants. External web search for close phrases found the Naor–Young Heisenberg endpoint results but no explicit general step-two Carnot extension. The result should therefore be treated as a likely new partial observation, pending expert literature review.

References

- [1] S.-Y. Ryoo. Quantitative nonembeddability of groups of polynomial growth into uniformly convex spaces. arXiv:2207.11305, version 2, 2025.
- [2] A. Naor and R. Young. Foliated corona decompositions. arXiv:2004.12522; *Acta Mathematica*, 229, 55–200, 2022.